

EE8203 - Design and Management of Data Networks

Department-Level Campus Network Design &
Management

Group Project - Design Document

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Version: 2.0

1. Department Design

This document presents the network design for the Faculty of Engineering at the University of Ruhuna, Hapugala premises. The design interconnects four academic and administrative departments:

Code	Department	Role
DEIE	Dept. of Electrical & Information Engineering	Engineering research and undergraduate labs
DCEE	Dept. of Civil & Environmental Engineering	Administration and project coordination offices
DMME	Dept. of Mechanical & Manufacturing Engineering	Workshop lab and server access zone
DIS	Dept. of Inter Disciplinary Studies	Faculty IT hub - hosts central server farm

The network follows a **three-layer hierarchical design** (Core, Distribution, Access) with VLAN segmentation, OSPF routing, NAT, and strict ACL-based security policies. The design supports automation via Python/Netmiko and Ansible, and monitoring via Zabbix.

2. Network Design Overview

2.1 Design Philosophy

The network is designed following industry best practices:

- **Hierarchical Design:** Three-layer model (Core, Distribution, Access) for scalability and fault isolation
- **VLAN Segmentation:** Separate VLANs for each department and management
- **Redundancy:** Loopback interfaces for OSPF stability
- **Security:** ACLs enforce inter-department access policies
- **Automation:** Infrastructure supports Netmiko and Ansible for configuration management
- **Monitoring:** Zabbix with SNMP provides visibility into network health

2.2 Network Hierarchy

Layer	Devices	Function
Core	R-CORE, R-EDGE, SW-CORE	High-speed backbone, WAN connectivity.
Distribution	SW-D-DEIE, SW-D-DCEE, SW-D-DMME	Aggregation, inter-VLAN routing (department gateways), policy enforcement.
Access	SW-A-DEIE, SW-A-DCEE, SW-A-DMME, SW-A-DIS	End-user connectivity, port security

3. IP Addressing Plan

3.1 Addressing Scheme Overview

The network uses private IP address space (10.0.0.0/8) with subnetting based on departmental requirements.

3.2 VLAN Subnets

VLAN ID	VLAN Name	Department	Subnet	Gateway	Purpose
10	VLAN_DEIE	DEIE	10.10.10.0/24	10.10.10.1	Engineering workstations
20	VLAN_DCEE	DCEE	10.10.20.0/24	10.10.20.1	Civil/Environmental dept hosts
30	VLAN_DMME	DMME	10.10.30.0/24	10.10.30.1	Mechanical/Manufacturing hosts
40	VLAN_DIS	DIS	10.10.40.0/24	10.10.40.1	Faculty IT hub / server zone
99	VLAN_MGMT	All devices	10.99.99.0/24	10.99.99.1	Out-of-band management (SSH only)
100	NATIVE	Trunk links	N/A	N/A	Native VLAN (untagged frames)

3.3 Point-to-Point Links

Link	Network	R-EDGE IP	R-CORE IP	SW-CORE IP
R-EDGE ↔ ISP-Router	200.200.200.0/30	200.200.200.2	-	-
R-EDGE ↔ R-CORE	10.255.0.0/30	10.255.0.1	10.255.0.2	-
R-CORE ↔ SW-CORE	10.255.0.4/30	-	10.255.0.5	10.255.0.6
SW-CORE ↔ SW-D-DEIE	10.255.0.8/30		10.255.0.9	10.255.0.10
SW-CORE ↔ SW-D-DCEE	10.255.0.12/30		10.255.0.13	10.255.0.14
SW-CORE ↔ SW-D-DMME	10.255.0.16/30		10.255.0.17	10.255.0.18

3.4 Management IP Addresses

Device	Management IP	VLAN	Interface
SW-CORE	10.99.99.1	99	Vlan99
SW-D-DEIE	10.99.99.11	99	Vlan99
SW-A-DEIE	10.99.99.21	99	Vlan99
SW-D-DCEE	10.99.99.12	99	Vlan99
SW-A-DCEE	10.99.99.22	99	Vlan99
SW-D-DMME	10.99.99.13	99	Vlan99
SW-A-DMME	10.99.99.23	99	Vlan99
SW-A-DIS	10.99.99.24	99	Vlan99
R-EDGE	10.255.0.1	-	Serial0/1/0
R-CORE	10.255.0.5	-	Gig0/0/0

3.5 End-Device IP Assignments

Device	VLAN	Static IP Address	Subnet Mask	Gateway
DEIE-PC1	10	10.10.10.10	255.255.255.0	10.10.10.1
DEIE-PC2	10	10.10.10.11	255.255.255.0	10.10.10.1
DCEE-PC1	20	10.10.20.10	255.255.255.0	10.10.20.1
DCEE-PC2	20	10.10.20.11	255.255.255.0	10.10.20.1
DMME-PC1	30	10.10.30.10	255.255.255.0	10.10.30.1
DMME-PC2	30	10.10.30.11	255.255.255.0	10.10.30.1
DIS-PC1	40	10.10.40.10	255.255.255.0	10.10.40.1
DIS-PC2	40	10.10.40.11	255.255.255.0	10.10.40.1
*DHCP/WEB Server	40	10.10.40.2	255.255.255.0	10.10.40.1
*DNS Server	40	10.10.40.3	255.255.255.0	10.10.40.1
*Google Server	-	8.8.8.8	255.255.255.0	8.8.8.1

Later Configured DHCP Server then got dynamic ip addresses for the vPCs

4. VLAN Scheme

4.1 VLAN Allocation

VLAN ID	Name	Type	Purpose
10	VLAN_DEIE	User	Department of Electrical & Information Engineering
20	VLAN_DCEE	User	Department of Civil & Environmental Engineering
30	VLAN_DMME	User	Department of Mechanical & Manufacturing Engineering
40	VLAN_DIS	Server	Department of Inter Disciplinary Studies / Server Farm
99	VLAN_MGMT	Management	Out-of-band management network
100	NATIVE	Native	Untagged frames on trunk links

4.2 VLAN Trunking

All inter-switch links are configured as **802.1Q trunks** with (MLSW)

- **Native VLAN:** 100 (untagged frames)
- **Allowed VLANs:** Department VLAN + Management VLAN + Native VLAN
- **Encapsulation:** dot1Q (default on Cisco switches) (Routers and other L2 switches)

Trunk Allowed VLANs

Link	Allowed VLANs
SW-CORE ↔ SW-A-DIS	40, 99, 100
SW-D-DEIE ↔ SW-A-DEIE	10, 99, 100
SW-D-DCEE ↔ SW-A-DCEE	20, 99, 100
SW-D-DMME ↔ SW-A-DMME	30, 99, 100

5. Topology Diagram

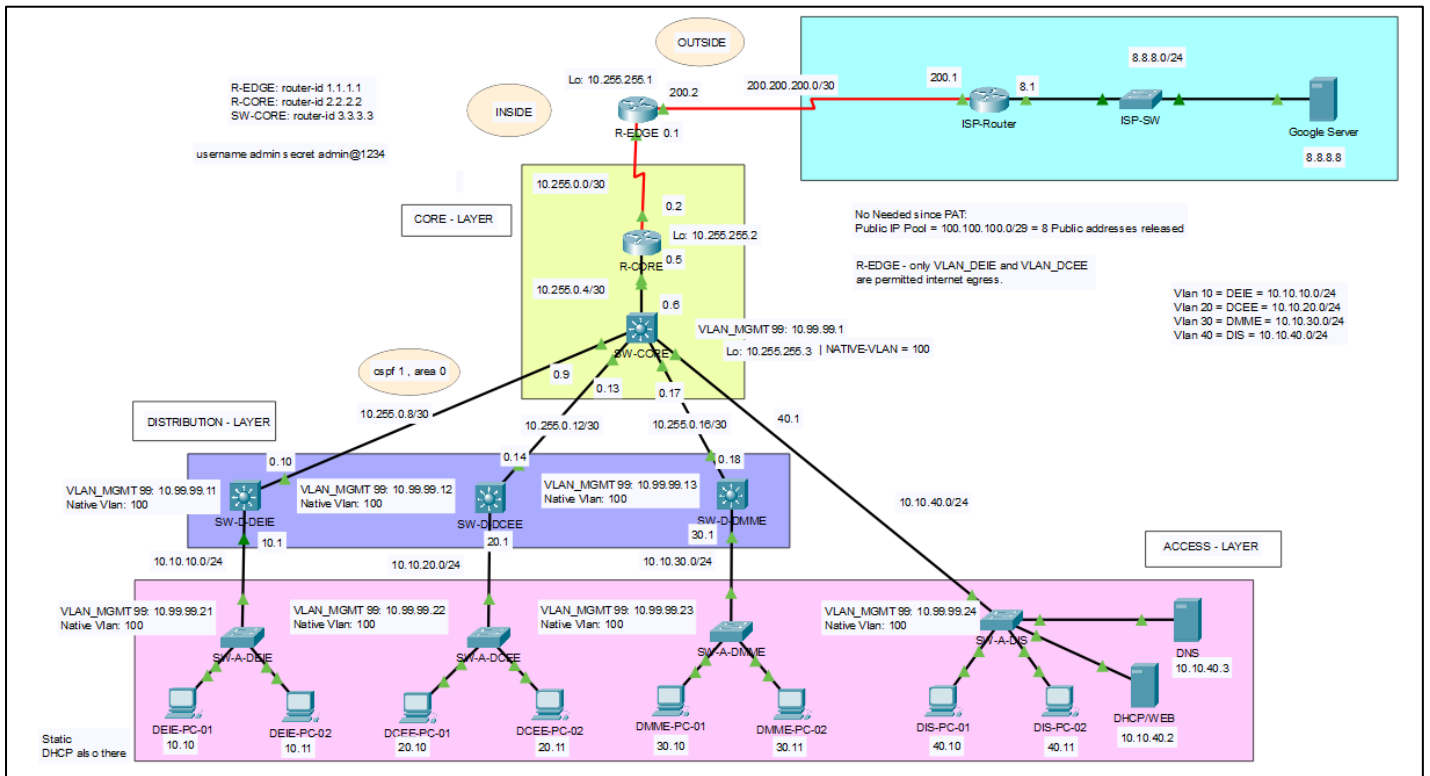


Figure 1: Topology Diagram V-2

Note: Here we used ISP-Router and ISP-Switch to test the connectivity with the google server in the OUTSIDE/WAN side

6. Device Selection Justification

6.1 Device Inventory (GNS3)

Device / Hostname	Qty	Image / Type	Primary Role
R-CORE	1	IOSv	Core router, OSPF area 0, NAT
R-EDGE	1	IOSv	Edge / WAN router, default route, NAT overload
SW-CORE	1	IOSvL2	Core L3 switch, inter-VLAN routing backbone
SW-D-DEIE, SW-D-DCEE, SW-D-DMME	3	IOSvL2	Distribution L3 switches
SW-A-DEIE, SW-A-DCEE, SW-A-DMME, SW-A-DIS	4	IOSvL2	Access L2 switches
VM-AUTO	1	Linux (Ubuntu)	Automation host - Python/Netmiko + Ansible + SSH
VM-ZABBIX	1	Linux (Ubuntu)	Zabbix server, SNMPv2c monitoring
VM-DHCP/WEB	1	Linux	DHCP/DNS server; optional web server
VPC end-hosts	8+	GNS3 VPCs	Two per department - used for ACL policy testing

6.2 Device Selection Rationale

Core Layer - R-CORE and R-EDGE (IOSv)

- OSPF Support: Full OSPF implementation required for dynamic routing
- NAT Capability: Supports NAT overload (PAT) for internet egress
- Scalability: Handles routing for the entire campus network

Why Cisco IOSv?

- IOSv is lightweight and runs efficiently in virtual environments
- Provides full enterprise routing features needed for the design
- Familiar CLI reduces implementation errors

Core Switch - SW-CORE (Cisco IOSvL2)

- Layer 3 Switching: Required for inter-VLAN routing at wire speed
- OSPF Routing: Participates in OSPF area 0 for dynamic routing
- VLAN Support: Supports VLAN creation, trunking, and SVIs
- Performance: Handles traffic between all four departments efficiently

Distribution Switches - SW-D (Cisco IOSvL2)

- Aggregation: Aggregates access layer switches for each department.
- Routing & SVIs: Acts as the default gateway and routes traffic for departmental VLANs via OSPF.
- Management: Each distribution switch has a management IP on VLAN 99.

Access Switches - SW-A (Cisco IOSvL2)

- End-User Connectivity: Provides access ports for end devices
- Port Security: Supports Portfast and BPDUguard for security
- VLAN Assignment: Assigns devices to appropriate departmental VLANs

Virtual Machines (Ubuntu)

- VM-AUTO: Linux is ideal for automation (Python, Netmiko, Ansible)
- VM-ZABBIX: Zabbix server runs natively on Linux
- VM-DHCP/WEB: Linux provides flexible DHCP and web server options

6.3 Tool Selection Rationale

Netmiko, Python for Routers

- Device-Specific: Handles Cisco IOS (prompts, paging)
- Flexibility: Python-based for custom logic and error handling
- Integration: Works well with inventory files (JSON/YAML)

Ansible for Switches

- Idempotency: Built-in functions prevent duplicate configs.
- Scalability: Designed for large infrastructure automation.
- Roles: Organizes configuration by function (VLANs, trunks, STP).

Why Different Tools for Different Tasks?

Aspect	Routers (Netmiko)	Switches (Ansible)
Task Complexity	Complex (NAT/PAT, OSPF, ACLs)	Structured (VLANs, trunking)
Scalability	Good for small-mid scale	Excellent for large scale
Learning Curve	Moderate	Steeper
Configuration	Python scripts	YAML playbooks

7. Network Architecture Details

7.1 Routing Protocol - OSPF

OSPF (Open Shortest Path First) is used as the dynamic routing protocol:

Parameter	Value
OSPF Process ID	1
Area	0 (Single Area)
Router IDs	1.1.1.1 (R-EDGE), 2.2.2.2 (R-CORE), 3.3.3.3 (SW-CORE), 4.4.4.4 (SW-D-DEIE), 5.5.5.5 (SW-D-DCEE), 6.6.6.6 (SW-D-DMME)

OSPF Network Statements:

Device	Network	Wildcard Mask	Area
R-EDGE	10.255.0.0	0.0.0.3	0
R-EDGE	10.255.255.1	0.0.0.0	0
R-CORE	10.255.0.0	0.0.0.3	0
R-CORE	10.255.0.4	0.0.0.3	0
R-CORE	10.255.255.2	0.0.0.0	0
SW-CORE	10.255.0.4	0.0.0.3	0
SW-CORE	10.255.0.8	0.0.0.3	0
SW-CORE	10.255.0.12	0.0.0.3	0
SW-CORE	10.255.0.16	0.0.0.3	0
SW-CORE	10.10.40.0	0.0.0.255	0
SW-CORE	10.99.99.0	0.0.0.255	0
SW-CORE	10.255.255.3	0.0.0.0	0
SW-D-DEIE	10.255.0.8	0.0.0.3	0
SW-D-DEIE	10.10.10.0	0.0.0.255	0
SW-D-DEIE	10.99.99.0	0.0.0.255	0
SW-D-DEIE	4.4.4.4	0.0.0.0	0
SW-D-DCEE	10.255.0.12	0.0.0.3	0
SW-D-DCEE	10.10.20.0	0.0.0.255	0
SW-D-DCEE	10.99.99.0	0.0.0.255	0
SW-D-DCEE	5.5.5.5	0.0.0.0	0
SW-D-DMME	10.255.0.16	0.0.0.3	0
SW-D-DMME	10.10.30.0	0.0.0.255	0
SW-D-DMME	10.99.99.0	0.0.0.255	0
SW-D-DMME	6.6.6.6	0.0.0.0	0

7.2 NAT Configuration

Component	Configuration
Inside Interfaces	Serial0/1/0 (R-EDGE to R-CORE)
Outside Interface	Serial0/1/1 (R-EDGE to ISP)
NAT ACL	Standard ACL 1 permits 10.10.10.0/24 and 10.10.20.0/24
NAT Type	PAT (Port Address Translation) / Overload
Translation	ip nat inside source list 1 interface Serial0/1/1 overload

